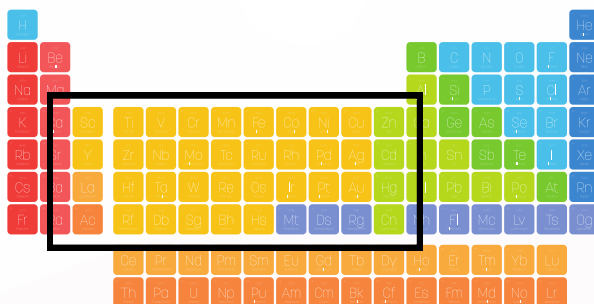


JEE (MAIN+ADVANCED)

Inorganic Chemistry

ENTHUSIAST COURSE



The image shows a periodic table of elements. A black rectangular box highlights the d-block, which consists of the transition metals. The elements included in the d-block are: Scandium (Sc), Titanium (Ti), Vanadium (V), Chromium (Cr), Manganese (Mn), Iron (Fe), Cobalt (Co), Nickel (Ni), Copper (Cu), and Zinc (Zn) in the first row; Yttrium (Y), Zirconium (Zr), Niobium (Nb), Molybdenum (Mo), Technetium (Tc), Ruthenium (Ru), Rhodium (Rh), Palladium (Pd), Silver (Ag), and Cadmium (Cd) in the second row; Lanthanum (La), Cerium (Ce), Praseodymium (Pr), Neodymium (Nd), Promethium (Pm), Samarium (Sm), Europium (Eu), Gadolinium (Gd), Terbium (Tb), Dysprosium (Dy), Holmium (Ho), Erbium (Er), Thulium (Tm), Ytterbium (Yb), and Lutetium (Lu) in the third row; and Hafnium (Hf), Tantalum (Ta), Tungsten (W), Rhenium (Re), Osmium (Os), Iridium (Ir), Platinum (Pt), Gold (Au), Mercury (Hg), and Thallium (Tl) in the fourth row.

STUDY MATERIAL

d- Block

English Medium

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d-BLOCK**TRANSITION ELEMENTS**

Definition : They are often called 'transition elements' because their position in the periodic table is between s-block and p-block elements

Typically, the transition elements have incompletely filled d-level. Since Zn group has d^{10} configuration in their ground state as well as in stable oxidation state, they are not considered as transition elements but they are d-block elements.

1st series										
	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Z	21	22	23	24	25	26	27	28	29	30
4s	2	2	2	1	2	2	2	2	1	2
3d	1	2	3	5	5	6	7	8	10	10

2nd series										
	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Z	39	40	41	42	43	44	45	46	47	48
5s	2	2	1	1	2	1	1	0	1	2
4d	1	2	4	5	5	7	8	10	10	10

3rd series										
	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Z	57	72	73	74	75	76	77	78	79	80
6s	2	2	2	2	2	2	2	1	1	2
5d	1	2	3	4	5	6	7	9	10	10

4th series										
	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Uub
Z	89	104	105	106	107	108	109	110	111	112
7s	2	2	2	2	2	2	2	2	1	2
6d	1	2	3	4	5	6	7	8	10	10

General Characteristics :

(i) **Metallic character** : They are all metal and good conductor of heat & electricity

(ii) **Electronic configuration** : $(n-1)d^{1-10}ns^{1-2}$

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
others are as usual	{			4s ¹					4s ¹	
				3d ⁵					3d ¹⁰	

(iii) M.P. Cr } $\longrightarrow$ Maximum
 Mo } 6 no. of unpaired e^- s
 W } are involved in metallic bonding
 Zn } lowest m.p.
 Cd } due to no unpaired e^-
 Hg } for metallic bonding

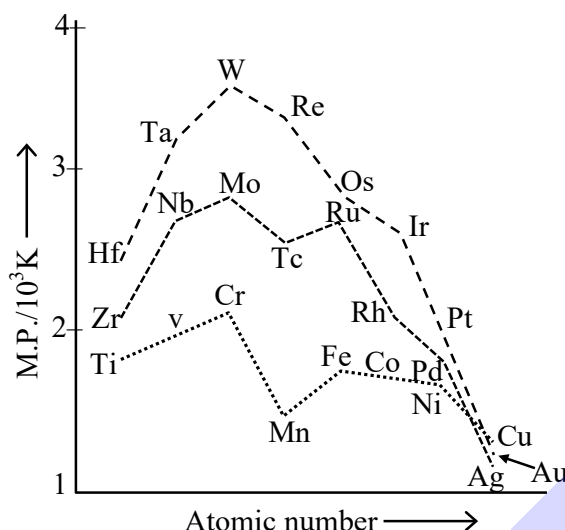


Fig. : Trends in melting points of transition elements

(iv) Variation in atomic radius :

Sc $\longrightarrow$ Mn Fe Co Ni Cu Zn
 decreases remains same increases again

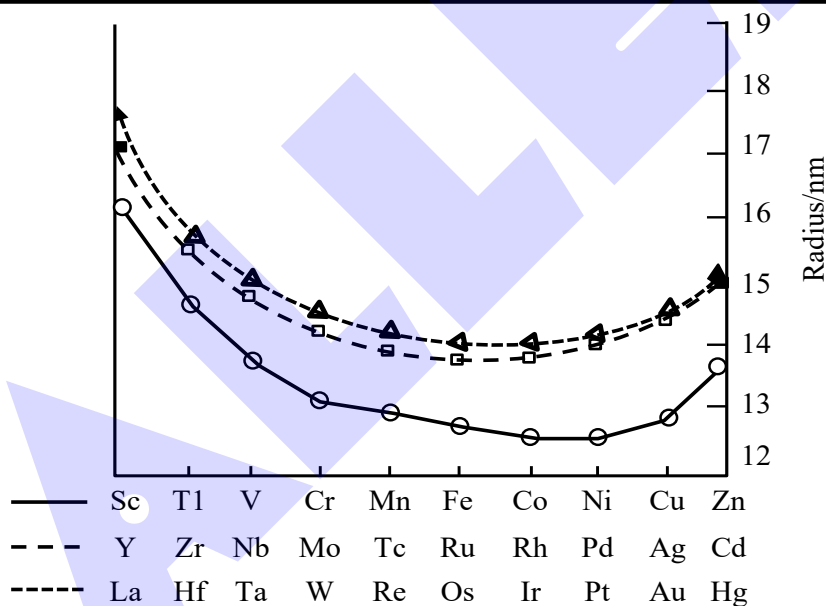


Fig. 8.3 : Trends in atomic radii of transition elements

(v) Ionisation energy :

1^{st} , 2^{nd} , 3^{rd} IE's are increasing from left to right for 1^{st} Transition series, but not regularly.
 For 2^{nd} IE $\text{Cr} > \text{Fe} > \text{Mn}$ and $\text{Cu} > \text{Zn}$
 For 3^{rd} IE $\text{Mn} > \text{Cr} > \text{Fe}$ and Zn has highest.

(vi) DENSITY

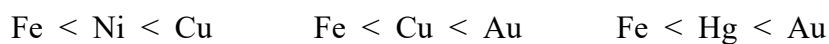
- The atomic volume of the transition elements are low, compared with s-block, so their density is comparatively high ($D = M/V$)
- Os (22.57 gm cm^{-3}) and Ir (22.61 gm cm^{-3}) have highest density.
- In all the groups (except IIIB) there is normal increase in density from 3d to 4d series, and from 4d to 5d, it increases just double. Due to lanthanide contraction
Ex. : $\text{Ti} < \text{Zr} \ll \text{Hf}$

(d) In 3d series

	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Density/g cm ⁻³	3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

(e) In 3d series highest density – Cu lowest density – Sc

(f) Some important orders of density


Electronic configurations and some other properties of the first series of transition elements

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Atomic number	21	22	23	24	25	26	27	28	29	30
Electronic configuration										
M	3d ¹ 4s ²	3d ² 4s ²	3d ³ 4s ²	3d ⁵ 4s ²	3d ⁵ 4s ²	3d ⁶ 4s ²	3d ⁷ 4s ²	3d ¹⁰ 4s ²	3d ¹⁰ 4s ¹	3d ¹⁰ 4s ²
M ⁺	3d ¹ 4s ¹	3d ² 4s ¹	3d ³ 4s ¹	3d ⁵	3d ⁵ 4s ¹	3d ⁶ 4s ¹	3d ⁷ 4s ¹	3d ⁸ 4s ¹	3d ¹⁰	3d ¹⁰ 4s ¹
M ²⁺	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	3d ⁸	3d ⁹	3d ¹⁰
M ³⁺	[Ar]	3d ¹	3d ²	3d ³	3d ⁴	3d ⁵	3d ⁶	3d ⁷	—	—
Enthalpy of atomisation, Δ _a H ⁰ /kJ mol ⁻¹	326	473	515	397	281	416	425	430	339	126
Ionisation Enthalpy, Δ _i H ⁰ /kJ mol ⁻¹										
Δ _i H ⁰ I	631	656	650	653	717	762	758	736	745	906
II	1235	1309	1414	1592	1509	1561	1644	1752	1958	1734
III	2393	2657	2833	2990	3260	2962	3243	3402	3556	3829
Metallic/ M	164	147	135	129	137	126	125	125	128	137
Ionic radii/pm										
M ²⁺	-	-	79	82	82	77	74	70	73	75
M ³⁺	73	67	64	62	65	65	61	60	-	-
Standard electrode potential, E ⁰ /V										
M ⁺² /M	-	-1.63	-1.18	-0.90	-1.18	-0.44	-0.28	-0.25	+0.34	-0.76
M ⁺³ /M ²⁺	-	-0.37	-0.26	-0.41	+1.57	+0.77	+1.97	-	-	-
Density/g cm ⁻³	3.43	4.1	6.07	7.19	7.21	7.8	8.7	8.9	8.9	7.1

VARIABLE OXIDATION STATES POSSIBLE :

- (1) The elements which give the greatest number of oxidation states occur in or near the middle of the series. Manganese, for example, exhibits all the oxidation states from +2 to +7.
- (2) The lesser number of oxidation states at the extreme ends stems from either too few electrons to lose or share (Sc, Ti) or too many d electrons (hence fewer orbitals available in which to share electrons with others) for higher valence (Cu, Zn).
- (3) Thus, early in the series scandium(II) is virtually unknown and titanium (IV) is more stable than Ti(III) or Ti(II).
- (4) At the other end, the only oxidation state of zinc is +2 (no d electrons are involved).
- (5) The maximum oxidation states of reasonable stability correspond in value to the sum of the s and d electrons upto manganese ($\text{Ti}^{\text{IV}}\text{O}_2$, $\text{V}^{\text{V}}\text{O}_2^+$, $\text{Cr}^{\text{VI}}\text{O}_4^{2-}$, $\text{Mn}^{\text{VII}}\text{O}_4^-$) followed by a rather abrupt decrease in stability of higher oxidation states, so that the typical species to follow are $\text{Fe}^{\text{II,III}}$, $\text{Co}^{\text{II,III}}$, Ni^{II} , $\text{Cu}^{\text{I,II}}$, Zn^{II} .
- (6) The variability of oxidation states, a characteristic of transition elements, arises out of incomplete filling of d orbitals in such a way that their oxidation states differ from each other by unity, e.g., V^{II} , V^{III} , V^{IV} , V^{V} .
- (7) This is in contrast with the variability of oxidation states of non transition elements where oxidation states normally differ by a unit of two.
- (8) An interesting feature in the variability of oxidation states of the d-block elements is noticed among the groups (groups 4 through 10).
- (9) In group 6, Mo(VI) and W(VI) are found to be more stable than Cr(VI). Thus Cr(VI) in the form of dichromate in acidic medium is a strong oxidising agent, whereas MoO_3 and WO_3 are not.
- (10) Low oxidation states are found when a complex compound has ligands capable of π -acceptor character in addition to the σ -bonding. For example, in $\text{Ni}(\text{CO})_4$ and $\text{Fe}(\text{CO})_5$, the oxidation state of nickel and iron is zero.
- (11) As the oxidation number of a metal increases, ionic character decreases. In the case of Mn, Mn_2O_7 is a covalent green oil. Even CrO_3 and V_2O_5 have low melting points. In these higher oxides, the acidic character is predominant. Thus, Mn_2O_7 gives HMnO_4 and CrO_3 gives H_2CrO_4 and $\text{H}_2\text{Cr}_2\text{O}_7$. V_2O_5 is, however, amphoteric though mainly acidic and it gives VO_4^{3-} as well as VO_2^+ salts. In vanadium there is gradual change from the basic V_2O_3 to less basic V_2O_4 and to amphoteric V_2O_5 . V_2O_4 dissolves in acids to give VO^{2+} salts. Similarly, V_2O_5 reacts with alkalis as well as acids to give VO_4^{3-} and VO_2^+ respectively. The well characterised CrO is basic but Cr_2O_3 is amphoteric.

Oxidation states of the 1st transition series

most common ones are in bold types :

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
			+1					+1	
	+2	+2	+2	+2	+2	+2	+2	+2	+2
+3	+3	+3	+3	+3	+3	+3	+3		
	+4	+4	+4	+4	+4	+4	+4		
		+5	+5	+5					
			+6	+6	+6				
				+7					

Trends in stability of higher oxidation state :

- (1) Table shows the stable halides of the 3d series of transition metals. The highest oxidation numbers are achieved in TiX_4 (tetrahalides), VF_5 and CrF_6 . The +7 state for Mn is not represented in simple halides but MnO_3F is known, and beyond Mn no metal has a trihalide except FeX_3 and CoF_3 .
- (2) The ability of fluorine to stabilise the highest oxidation state is due to either higher lattice energy as in the case of CoF_3 , or higher bond enthalpy terms for the higher covalent compounds, e.g., VF_5 and CrF_6 .
- (3) Although V^{V} is represented only by VF_5 , the other halides, however, undergo hydrolysis to give oxohalides, VOX_3 .
- (4) Another feature of fluorides is their instability in the low oxidation states e.g., VX_2 ($\text{X} = \text{Cl}, \text{Br}$ or I)

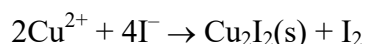
Formulas of halides of 3d-metals

Oxidation Number									
+6			CrF_6						
+5		VF_5	CrF_5						
+4	TiX_4	VX_4^{I}	CrX_4	MnF_4					
+3	TiX_3	VX_3	CrX_3	MnF_3	FeX_3^{I}	CoF_3			
+2	$\text{TiX}_2^{\text{III}}$	VX_2	CrX_2	MnX_2	FeX_2	CoX_2	NiX_2	CuX_2^{I}	ZnX_2
+1								CuX^{III}	

Key : $\text{X} = \text{F} \rightarrow \text{I}$; $\text{X}^{\text{I}} = \text{F} \rightarrow \text{Br}$; $\text{X}^{\text{II}} = \text{F} \rightarrow \text{Cl}$; $\text{X}^{\text{III}} = \text{Cl} \rightarrow \text{I}$

and the same applies to CuX . On the other hand, all Cu(II) halides are known except the iodide.

In this case, Cu^{2+} oxidises I^- to I_2 :



- (5) However, many copper (I) compounds are unstable in aqueous solution and undergo disproportionation.



- (6) The stability of $\text{Cu}^{2+}(\text{aq})$ rather than $\text{Cu}^+(\text{aq})$ is due to the much more negative $\Delta_{\text{Hyd}}H^\circ$ of $\text{Cu}^{2+}(\text{aq})$ than Cu^+ , which more than compensates for the second ionisation enthalpy of Cu.
- (7) The ability of oxygen to stabilise the highest oxidation state is demonstrated in the oxides.
- (8) The highest oxidation number in the oxides coincides with the group number and is attained in Sc_2O_3 to Mn_2O_7 .
- (9) Beyond Group 7, no higher oxides of Fe above Fe_2O_3 , are known, although ferrates (VI)(FeO_4) $^{2-}$, are formed in alkaline media but they readily decompose to Fe_2O_3 and O_2 .
- (10) Besides the oxides, oxocations stabilise V^{V} as VO^+_{2-} , V^{IV} as VO^{2+} and Ti^{IV} as TiO^{2+} .
- (11) The ability of oxygen to stabilise these high oxidation states exceeds that of fluorine. Thus the highest Mn fluoride is MnF_4 whereas the highest oxide is Mn_2O_7 . The ability of oxygen to form multiple bonds to metals explains its superiority.
- (12) In the covalent oxide Mn_2O_7 , each Mn is tetrahedrally surrounded by O's including a Mn–O–Mn bridge.
- (13) The tetrahedral $[\text{MO}_4]^n$ ions are known for V^{V} , Cr^{VI} , Mn^{V} , Mn^{VI} and Mn^{VII} .

FORMATION OF COLOURED ION :

Colour : (aquated)

$\text{Ti}^{4+} \longrightarrow$ colourless

$\text{V}^{4+} \longrightarrow$ blue

$\text{V}^{2+} \longrightarrow$ violet

$\text{Cr}^{3+} \longrightarrow$ violet

$\text{Mn}^{2+} \longrightarrow$ light pink

$\text{Fe}^{3+} \longrightarrow$ yellow

$\text{Ni}^{2+} \longrightarrow$ green

$\text{Zn}^{2+} \longrightarrow$ colourless

$\text{Sc}^{3+} \longrightarrow$ colourless

$\text{Ti}^{3+} \longrightarrow$ purple

$\text{V}^{3+} \longrightarrow$ green

$\text{Cr}^{2+} \longrightarrow$ blue

$\text{Mn}^{3+} \longrightarrow$ violet

$\text{Fe}^{2+} \longrightarrow$ light green

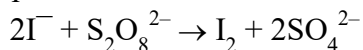
$\text{Co}^{2+} \longrightarrow$ pink

$\text{Cu}^{2+} \longrightarrow$ blue

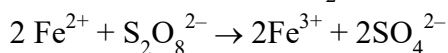
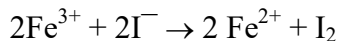
CATALYTIC PROPERTIES

- (1) The transition metals and their compounds are known for their catalytic activity. This activity is ascribed to their ability to adopt multiple oxidation states and to form complexes. Vanadium(V) oxide (in Contact Process), finely divided iron (in Haber's Process), and nickel (in Catalytic Hydrogenation) are some of the examples.
- (2) Catalysts at a solid surface involve the formation of bonds between reactant molecules and atoms of the surface of the catalyst (first row transition metals utilise 3d and 4s electrons for bonding).
- (3) This has the effect of increasing the concentration of the reactants at the catalyst surface and also weakening of the bonds in the reacting molecules (the activation energy is lowering).

- (4) Also because the transition metal ions can change their oxidation states, they become more effective as catalysts. For example, iron(III) catalyses the reaction between iodide and persulphate ions.



An explanation of this catalytic action can be given as:



Catalyst	Used
TiCl_3	→ Used as the Ziegler-Natta catalyst in the production of polythene.
V_2O_5	→ Convert SO_2 to SO_3 in the contact process for making H_2SO_4
MnO_2	→ Used as a catalyst to decompose KClO_3 to give O_2
Fe	→ Promoted iron is used in the Haber-Bosch process for making NH_3
FeCl_3	→ Used in the production of CCl_4 from CS_2 and Cl_2
PdCl_2	→ Wacker process for converting $\text{C}_2\text{H}_4 + \text{H}_2\text{O} + \text{PdCl}_2$ to $\text{CH}_3\text{CHO} + 2\text{HCl} + \text{Pd}$.
Pd	→ Used for hydrogenation (e.g. phenol to cyclohexanone).
Pt/PtO	→ Adams catalyst, used for reductions.
Pt	→ Formerly used for $\text{SO}_2 \rightarrow \text{SO}_3$ in the contact process for making H_2SO_4
Pt/Rh	→ Formerly used in the Ostwald process for making HNO_3 to oxidize NH_3 to NO
Cu	→ Is used in the direct process for manufacture of $(\text{CH}_3)_2\text{SiCl}_2$ used to make silicones.
Cu/V	→ Oxidation of cyclohexanol/cyclohexanone mixture to adipic acid which is used to make nylon-66
CuCl_2	→ Deacon process of making Cl_2 from HCl
Ni	→ Raney nickel, numerous reduction processes (e.g. manufacture of hexamethylenediamine, production of H_2 from NH_3 , reducing anthraquinone to anthraquinol in the production of H_2O_2)
$\text{FeSO}_4 + \text{H}_2\text{O}_2$	→ Used as Fenton's reagent for oxidizing alcohols to aldehydes.

Formation of Interstitial Compounds

Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals. The principal physical and chemical characteristics of these compounds are as follows :

- They have high melting points, higher than those of pure metals.
- They are very hard, some borides approach diamond in hardness.
- They retain metallic conductivity.
- They are chemically inert.

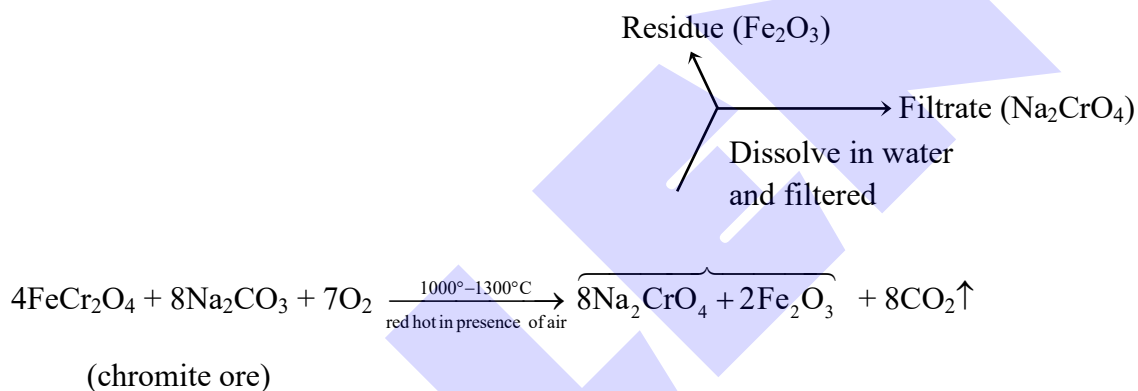
Alloy Formation

An alloy is a blend of metals prepared by mixing the components. Alloys may be homogeneous solid solutions in which the atoms of one metal are distributed randomly among the atoms of the other. Such alloys are formed by atoms with metallic radii that are within about 15 percent of each other. Because of similar radii and other characteristics of transition metals, alloys are readily formed by these metals. The alloys so formed are hard and have often high melting points.

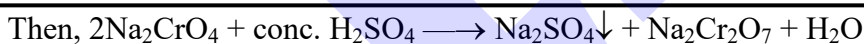
The best known are ferrous alloys: chromium, vanadium, tungsten, molybdenum and manganese are used for the production of a variety of steels and stainless steel. Alloys of transition metals with non transition metals such as brass (copper-zinc) and bronze (copper-tin), are also of considerable industrial importance.

CHROMATE –DICHROMATE

Preparation :



[Lime (CaO) added with Na_2CO_3 which keeps the mass porous so that air has access to all parts and prevents fusion]

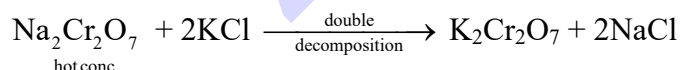


Its solubility
 upto 32°C increases
 and then decreases

Hence, suitable temp. is to be
 employed to crystallise out
 Na_2SO_4 first.

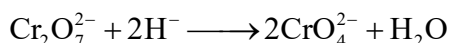
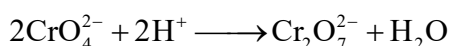
Then $\text{Na}_2\text{Cr}_2\text{O}_7$ is crystallised out as $\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2\text{H}_2\text{O}$ on evaporation.
 (red crystal)

How to get $\text{K}_2\text{Cr}_2\text{O}_7$:

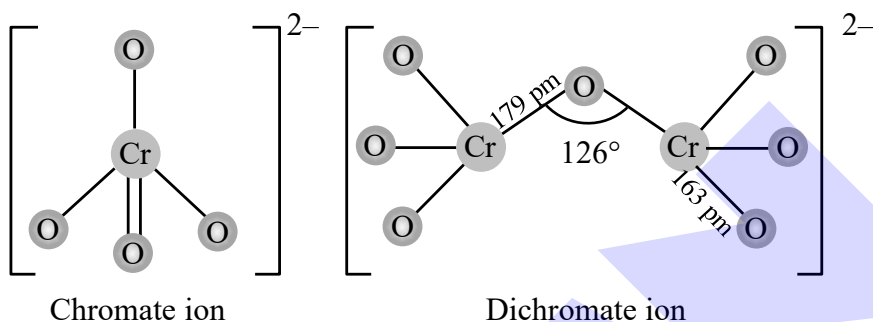
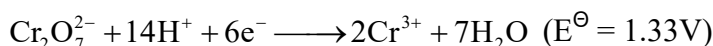


NaCl crystallises out first and filtered off. Then $\text{K}_2\text{Cr}_2\text{O}_7$ crystallised out on cooling

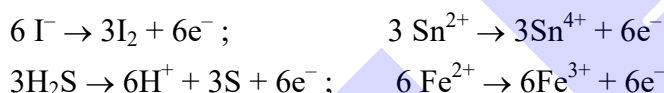
The chromates and dichromates are interconvertible in aqueous solution depending upon pH of the solution. The oxidation state of chromium in chromate and dichromate is the same.



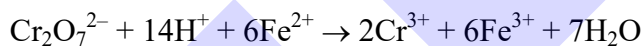
The structures of chromate ion, CrO_4^{2-} and the dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$ are shown below. The chromate ion is tetrahedral whereas the dichromate ion consists of two tetrahedra sharing one corner with Cr–O–Cr bond angle of 126° . Sodium and potassium dichromates are strong oxidising agents; the sodium salt has a greater solubility in water and is extensively used as an oxidising agent in organic chemistry. Potassium dichromate is used as a primary standard in volumetric analysis. In acidic solution, its oxidising action can be represented as follows:



Thus, acidified potassium dichromate will oxidise iodides to iodine, sulphides to sulphur, tin(II) to tin(IV) and iron(II) salts to iron(III). The half-reactions are noted below:

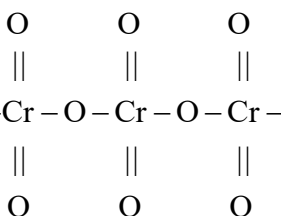


The full ionic equation may be obtained by adding the half-reaction for potassium dichromate to the half-reaction for the reducing agent, for e.g.,



* Similarities between hexavalent Cr & S-compounds :

- (i) SO_3 & $\text{CrO}_3 \longrightarrow$ both acidic.
- (ii) $\text{S} \longrightarrow \text{SO}_4^{2-}$, $\text{S}_2\text{O}_7^{2-}$, $\text{Cr} \longrightarrow \text{CrO}_4^{2-}$, $\text{Cr}_2\text{O}_7^{2-}$
- (iii) CrO_4^{2-} & SO_4^{2-} are isomorphous
- (iv) SO_2Cl_2 & $\text{CrO}_2\text{Cl}_2 \xrightarrow{\text{OH}^-} \text{SO}_4^{2-}$ & CrO_4^{2-} respectively.
- (v) SO_3Cl^- & $\text{CrO}_3\text{Cl}^- \xrightarrow{\text{OH}^-} \text{SO}_4^{2-}$ & CrO_4^{2-}
- (vi) CrO_3 & $\beta(\text{SO}_3)$ has same structure



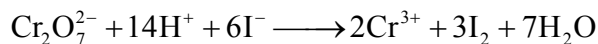
Q. In laboratory $\text{K}_2\text{Cr}_2\text{O}_7$ is used mainly not $\text{Na}_2\text{Cr}_2\text{O}_7$. Why?

Sol. $\text{Na}_2\text{Cr}_2\text{O}_7$ is deliquescent enough and changes its concentration and can not be taken as primary standard solution whereas $\text{K}_2\text{Cr}_2\text{O}_7$ has no water of crystallisation and not deliquescent.

Q. How to standardise $\text{Na}_2\text{S}_2\text{O}_3$ solution in iodometry?

Sol. $\text{K}_2\text{Cr}_2\text{O}_7$ is primary standard $\Rightarrow$ strength is known by weighing the salt in chemical balance and dissolving in measured amount of water.

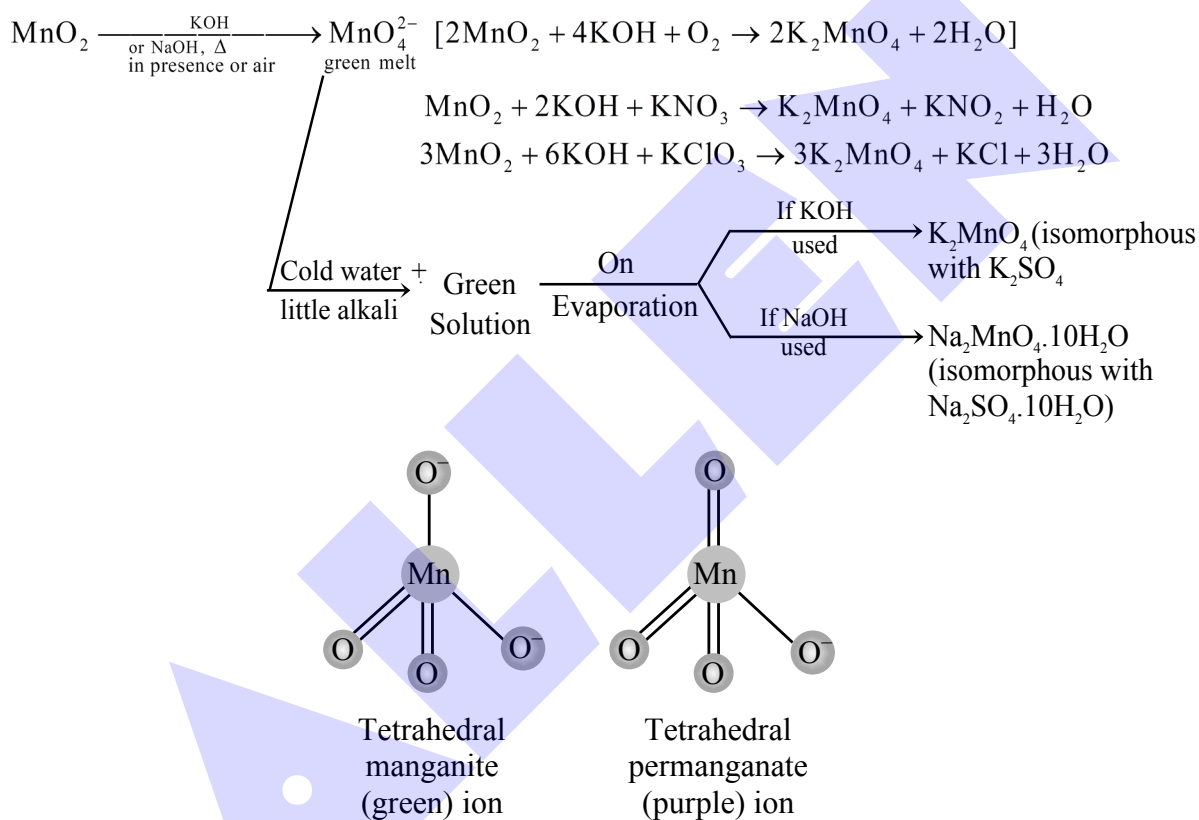
Then in acidic solution add. KI



This I_2 is liberated can be estimated with $\text{S}_2\text{O}_3^{2-}$.

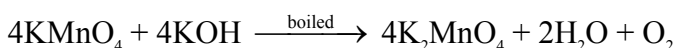
MANGANATE & PERMANGANATE

PREPARATION OF MANGANATE (MnO_4^{2-}) :-

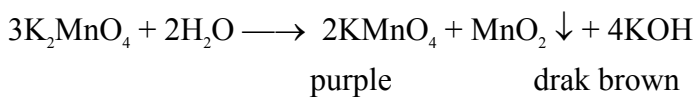


In presence of KClO_3 & KNO_3 the above reaction is more faster because these two on decomposition provides O_2 easily.

* Manganate is also obtained when KMnO_4 is boiled with KOH .

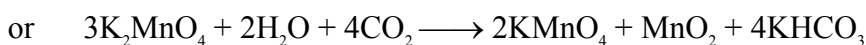


Properties : The above green solution is quite stable in alkali, but in pure water and in presence of acids, depositing MnO_2 and giving a purple solution of permanganate.



Prob. : $E^\circ_{\text{MnO}_4^{2-}/\text{MnO}_2} = 2.26 \text{ V}$; $E^\circ_{\text{MnO}_4^{2-}/\text{MnO}_4^-} = -0.56 \text{ V}$

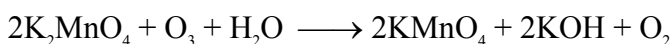
Prove that MnO_4^{2-} will disproportionate in acidic medium.

Conversion of MnO_4^{2-} to MnO_4^- 

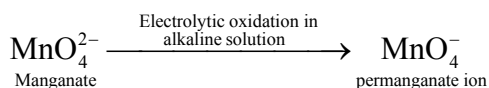
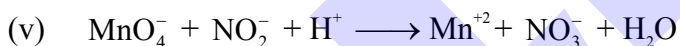
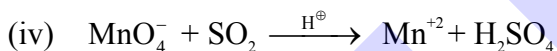
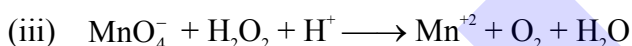
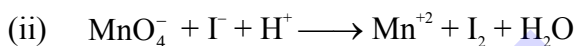
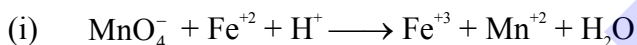
But in the above method $\frac{1}{3}$ of Mn is lost as MnO_2 but when oxidised either by Cl_2 or by O_3



OR

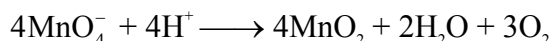


OR

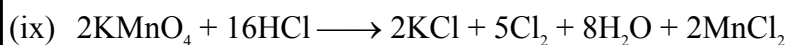
**Oxidising Prop. of KMnO_4 : (in acidic medium)**

(1)* It is not a primary standard since it is difficult to get it in a high degree of purity and free from traces of MnO_2 .

(2)* It is slowly reduced to MnO_2 especially in presence of light or acid



Hence it should be kept in dark bottles and standardise just before use.



Note: Permanganate titrations in presence of hydrochloric acid are unsatisfactory since hydrochloric acid is oxidised to chlorine.

Oxidising Prop. of KMnO_4 in neutral or faintly alkaline solution.

- (i) $2\text{KMnO}_4 + \text{H}_2\text{O} + \text{KI} \longrightarrow 2\text{MnO}_2 + 2\text{KOH} + \text{KIO}_3$
 (ii) $2\text{KMnO}_4 + 3\text{HCO}_2\text{K} \longrightarrow 2\text{MnO}_2 + \text{KHCO}_3 + 2\text{K}_2\text{CO}_3 + \text{H}_2\text{O}$
 (iii) $2\text{KMnO}_4 + 3\text{H}_2\text{O}_2 \longrightarrow 2\text{KOH} + 2\text{MnO}_2 + 2\text{H}_2\text{O} + 3\text{O}_2$
 (iv) Thiosulphate is oxidised almost quantitatively to sulphate:
 $8\text{MnO}_4^- + 3\text{S}_2\text{O}_3^{2-} + \text{H}_2\text{O} \longrightarrow 8\text{MnO}_2 + 6\text{SO}_4^{2-} + 2\text{OH}^-$

Oxidising Prop. in neutral or weakly acidic solution:

- (i) $2\text{KMnO}_4 + 3\text{MnSO}_4 + 2\text{H}_2\text{O} \xrightarrow{\text{in presence Zn}^{+2} \text{ or ZnO}} 5\text{MnO}_2 + \text{K}_2\text{SO}_4 + 2\text{H}_2\text{SO}_4$
 or $\text{MnO}_4^- + \text{Mn}^{+2} + 2\text{H}_2\text{O} \longrightarrow 5\text{MnO}_2 + 4\text{H}^+$

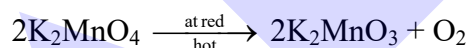
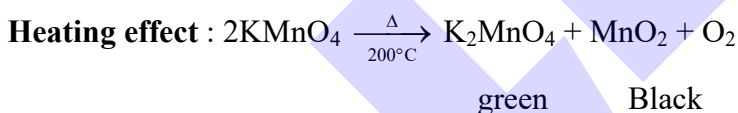
In absence of Zn^{+2} ions, some of the Mn^{+2} ion may escape, oxidation through the formation of insoluble $\text{Mn}^{\text{II}}[\text{Mn}^{\text{IV}}\text{O}_3]$ manganous permanganite.

Uses of KMnO_4 :

Besides its use in analytical chemistry, potassium permanganate is used as a favourite oxidant in preparative organic chemistry. Its uses for the bleaching of wool, cotton, silk and other textile fibres and for the decolourisation of oils are also dependent on its strong oxidising power.

** In laboratory conversion of Mn^{+2} to MnO_4^- is done by :

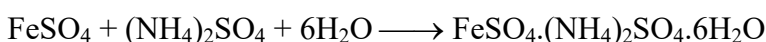
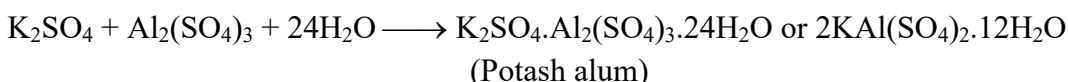
- (i) PbO_2 (ii) $\text{Pb}_3\text{O}_4 + \text{HNO}_3$ (iii) $\text{Pb}_2\text{O}_3 + \text{HNO}_3$ (iv) $\text{NaBiO}_3 / \text{H}^+$
 (v) $(\text{NH}_4)_2\text{S}_2\text{O}_8 / \text{H}^+$ (vi) $\text{KIO}_4 / \text{H}^+$

**SOME IMPORTANT EXPERIMENTS****Experiment 1****Aim**

To prepare double salts: ferrous ammonium sulphate (Mohr's salt) and potash alum.

Theory

When a mixture containing equimolar proportions of potassium sulphate and aluminium sulphate or ferrous sulphate and ammonium sulphate is crystallised from its solution, a double salt is formed. The formation of double salt may be shown as follows:



Ferrous ammonium sulphate (Mohr's salt)

Fe^{2+} and Al^{3+} ions undergo hydrolysis, therefore, while preparing aqueous solutions of ferrous sulphate and aluminium sulphate in water, 2-3 mL **dilute sulphuric acid is added to prevent the hydrolysis of these salts.**

Material Required

- | | | |
|-------------------------|-------|---------------------------------------|
| • Beaker (50 mL) | : One | |
| • Conical flask (50 mL) | : One | • Potassium sulphate : As per need |
| • Trough | : One | • Aluminium sulphate : As per need |
| • Glass rod | : One | • Ferrous sulphate : As per need |
| • Tripod stand | : One | • Ammonium sulphate : As per need |
| • Funnel | : One | • Dilute sulphuric acid : As per need |
| • Wire gauze | : One | • Ethanol : As per need |

Procedure**(a) Preparation of Double Salt: Potassium Aluminium Sulphate (Potash Alum)**

- Take 10 mL of distilled water in a 50 mL beaker and heat it to about 40°C. Dissolve 6.6 g of aluminium sulphate in it and add about 0.4 mL of dilute sulphuric acid.
- Weigh 2.4 g of powdered potassium sulphate and transfer it to the above solution.
- Heat the solution with constant stirring till potassium sulphate dissolves completely.
- Allow the solution to cool to room temperature slowly.
- On cooling, white crystals of potash alum separate out.
- Decant the mother liquor and wash the crystals by gently shaking with 1:1 cold water and alcohol mixture.
- Filter the crystals, dry these between the folds of a filter paper and note the yield.

(b) Preparation of Double Salt: Ferrous Ammonium Sulphate

- Dissolve 3.5 g of ferrous sulphate and 1.7 g of ammonium sulphate (weighed separately), in 5 mL of distilled water contained in a 50 mL conical flask by heating. Add about 0.5 mL of dilute sulphuric acid to the flask and concentrate the solution by heating till the crystallization point is reached.
- Allow the mixture to cool to room temperature slowly.
- On cooling, light green crystals of ferrous ammonium sulphate separate out.
- Decant the mother liquor and wash the crystals by shaking with very small amounts of 1:1 cold water and alcohol mixture to remove sticking mother liquor.
- Separate the crystals by filtration wash with alcohol, dry between the folds of a filter paper and record the yield.

Result

Yield of potash alum/Mohr's salt is _____ g.

Precautions

- Cool the solution slowly to get good crystals. Avoid rapid cooling.
- Do not disturb the solution while cooling.
- Avoid prolonged heating while preparing crystals of ferrous ammonium sulphate, as it may oxidise ferrous ions to ferric ions and change the stoichiometry of the crystals.

Discussion Questions

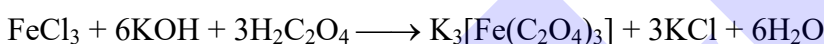
- (i) Why do we take equimolar quantities of reacting compounds in the preparation of double salts?
- (ii) In the preparation of ferrous ammonium sulphate, can concentrated sulphuric acid be used in place of dilute sulphuric acid? Explain.
- (iii) What is the difference between iron compounds; $K_4[Fe(CN)_6]$ and $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$?
- (iv) What is the action of heat on potash alum?
- (v) Why does an aqueous solution of potash alum turn blue litmus red?
- (vi) What are the isomorphous substances?
- (vii) Give the name of some of the alums where cations are other than Al^{+3} .
- (viii) What is the difference between a complex compound and a double salt?

Experiment 2**Aim**

To prepare potassium trioxalatoferrate(III).

Theory

When hydrated ferric chloride is dissolved in aqueous solution of oxalic acid containing potassium hydroxide, green crystals of potassium trioxalatoferrate(III) are obtained. The reaction involved in the formation of these green crystals is as follows:

**Material Required**

• Beaker (50 mL)	: One		
• Porcelain dish	: One	• Ferric chloride	: 2.5 g
• Water bath	: One	• Potassium hydroxide	: 3.8 g
• Glass rod	: One	• Oxalic acid	: 3.0 g
• Funnel	: One	• Ethanol	: As per need

Procedure

- (i) Prepare a solution of 3.0 g of oxalic acid in 12.5 mL of hot water contained in a clean 50 mL beaker.
- (ii) To the above solution, add 3.8 g of potassium hydroxide gradually in lots, with stirring so that it dissolves completely.
- (iii) Add 2.5 g of ferric chloride into the above solution with constant stirring till it is completely dissolved.
- (iv) Filter the solution and concentrate the green filtrate by heating in a porcelain dish over a water bath and cool the mixture slowly.
- (v) Filter the crystals so formed, wash with 1:1 mixture of cold water and alcohol and dry them by pressing between the folds of a filter paper.

Result

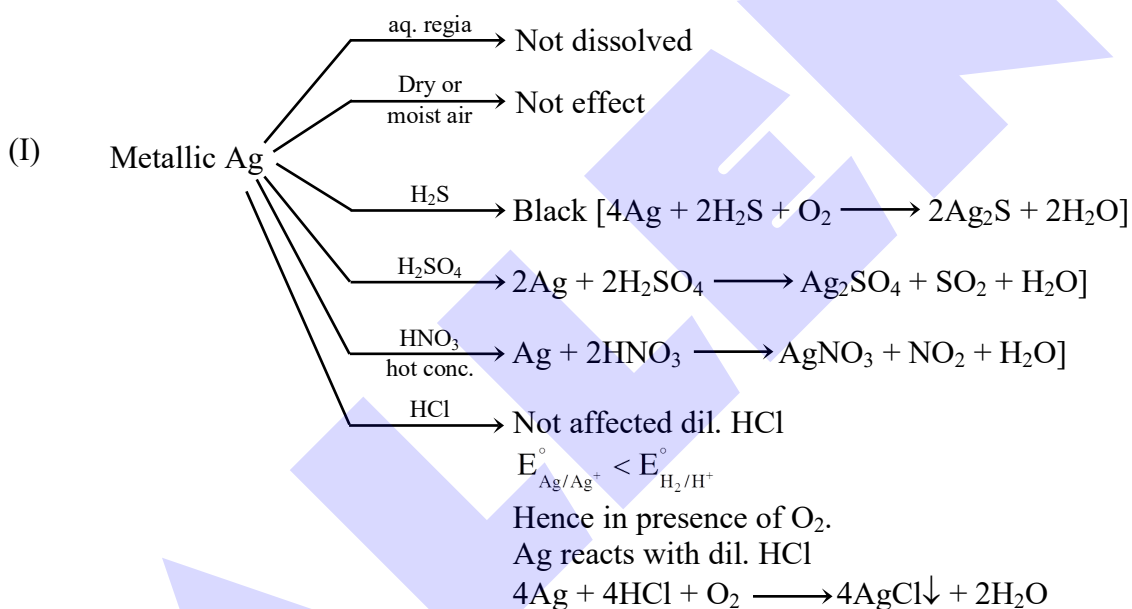
Yield of potassium trioxalatoferrate (III) is _____ g.

Precautions

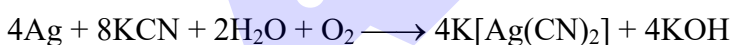
- Do not evaporate the entire solvent when the solution is being concentrated.
- Weigh the desired quantities of different substances accurately.
- Maintain the temperature of hot water around 40°C.
- Add potassium hydroxide to oxalic acid solution in small lots.

Discussion Questions

- Give IUPAC name of the compound called potassium ferrioxalate.
- What is the coordination number of iron in potassium trioxalatoferate(III)?
- Give two examples of bidentate ligands other than oxalate ion.
- Why does the compound, potassium trioxalatoferate(III), not give tests for ferric ions?
- What are chelates ?

SILVER AND ITS COMPOUND

In the same way in presence of O₂, Ag complexes with NaCN / KCN.

***AgNO₃***

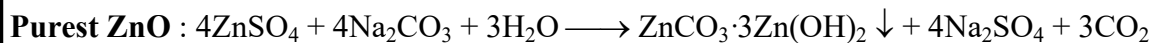
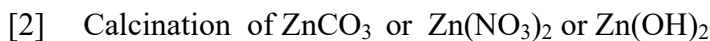
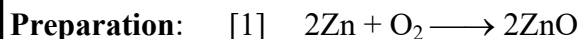
Preparation : Reaction of Ag with dilute HNO₃ or conc. HNO₃.

Properties :

- It is called as lunar caustic because in contact with skin it produces burning sensation like that of caustic soda with the formation of finely divided silver (black colour)
- Thermal decomposition : $2\text{AgNO}_3(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{O}_2(\text{g}) + 2\text{NO}_2(\text{g})$
- Props. of AgNO₃
 $6\text{AgNO}_3 + 3\text{I}_2 + 3\text{H}_2\text{O} \longrightarrow 5\text{AgI} + \text{AgIO}_3 + 6\text{HNO}_3$
 (excess)

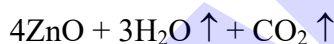
ZINC COMPOUNDS**ZnO**

It is called as phillospher's wool due to its wooly flock type appearance



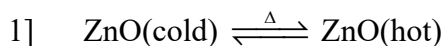
white basic zinc

carbonate



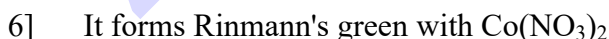
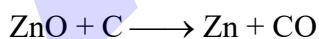
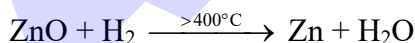
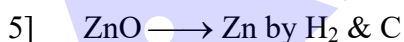
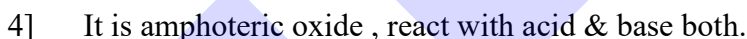
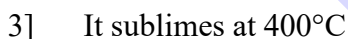
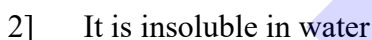
pure

Properties:



white

yellow



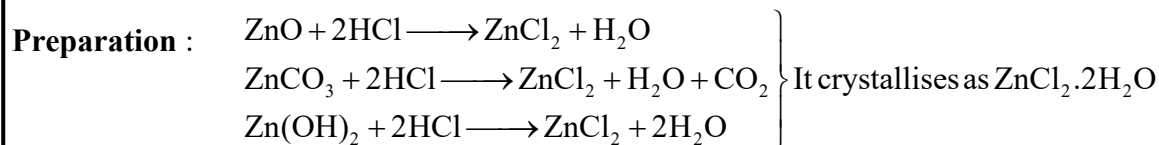
Rinmann's green

Uses:

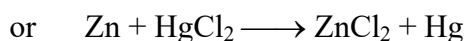
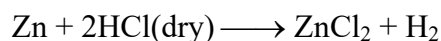
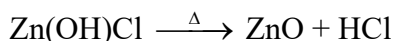
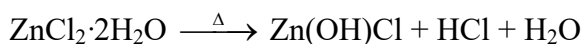
(1) As white pigment, it is superior than white lead because it does not turn into black

(2) Rinmann's green is used as green pigment

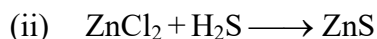
(3) It is used as zinc ointment in medicine

ZnCl₂

Anh. ZnCl₂ cannot be made by heating ZnCl₂·2H₂O because



Properties: (i) It is deliquescent white solid (when anhydrous)



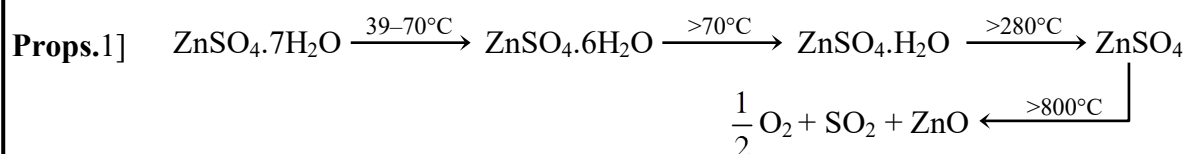
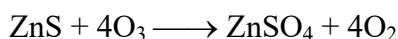
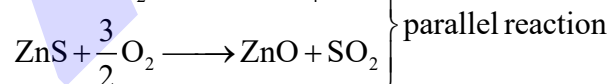
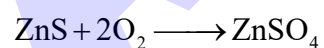
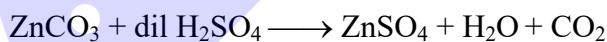
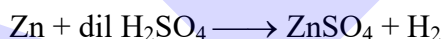
Uses: 1] Used for impregnating timber to prevent destruction by insects

2] As dehydrating agent when anhydrous

3] ZnO·ZnCl₂ used in dental filling

ZnSO₄

Preparation:→



Uses: 1] in eye lotion

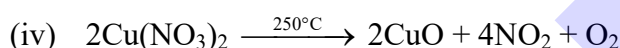
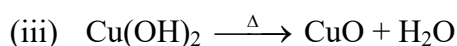
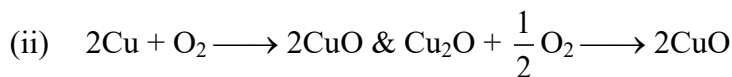
2] Lithophone (ZnS + BaSO₄) is used as white pigment

COPPER COMPOUNDS**CuO**

Preparation : (i) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2 \xrightarrow{\Delta} 2\text{CuO} + \text{H}_2\text{O} + \text{CO}_2$ (Commercial process)

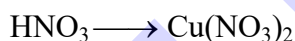
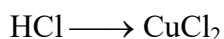
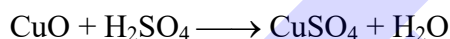
Malachite Green

(native Cu-carbonate)

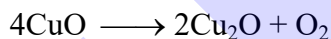


Properties : (i) CuO is insoluble in water

(ii) Readily dissolves in dil. acids



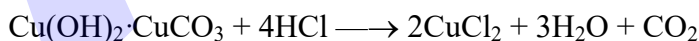
(iii) It decomposes when, heated above 1100°C



(iv) CuO is reduced to Cu by H_2 or C under hot condition

**CuCl₂**

Preparation : $\text{CuO} + 2\text{HCl}(\text{conc.}) \longrightarrow \text{CuCl}_2 + \text{H}_2\text{O}$



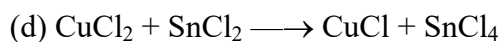
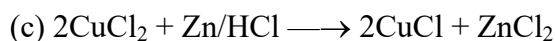
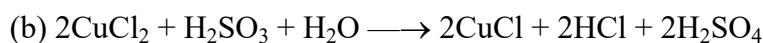
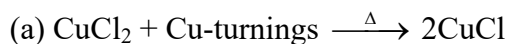
Properties : (i) It is crystallised as $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ of Emerald green colour

(ii) dil. solution in water is blue in colour due to formation of $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ complex.

(iii) When conc. HCl or KCl added to dil. solution of CuCl_2 the colour changes into yellow, owing to the formation of $[\text{CuCl}_4]^{2-}$

(iv) The conc. aq. solution is green in colour having the two complex ions in equilibrium $2[\text{Cu}(\text{H}_2\text{O})_4]\text{Cl}_2 \rightleftharpoons [\text{Cu}(\text{H}_2\text{O})_4]^{2+} + [\text{CuCl}_4]^{2-} + 4\text{H}_2\text{O}$

(v) $\text{CuCl}_2 \longrightarrow \text{CuCl}$ by no. of reagents



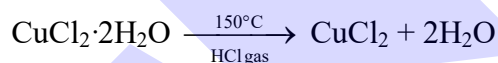
** $\text{CuF}_2 \cdot 2\text{H}_2\text{O} \longrightarrow$ light blue

$\text{CuCl}_2 \cdot 2\text{H}_2\text{O} \longrightarrow$ green

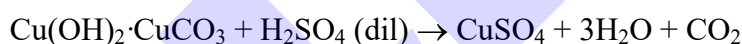
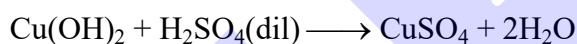
$\text{CuBr}_2 \longrightarrow$ almost black

CuI_2 does not exist

{ Anh. CuCl_2 is dark brown mass obtained
by heating $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$ at 150°C in presence
of HCl vap.



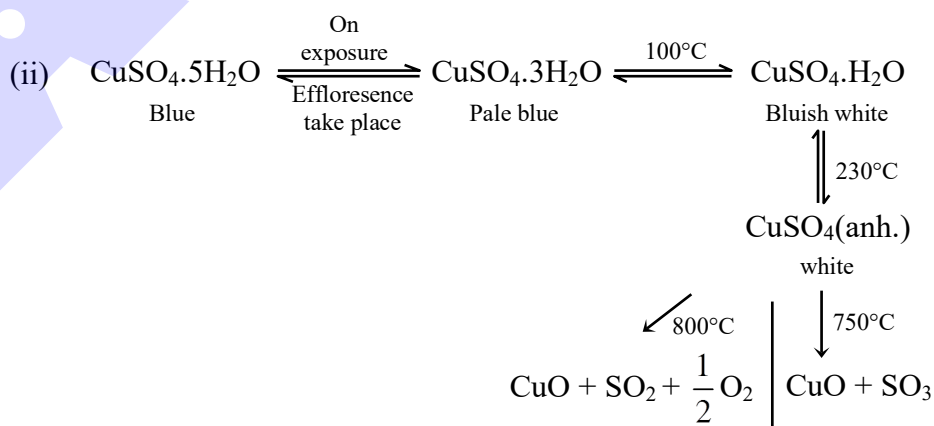
CuSO_4



(Scrap)

$\text{Cu} + \text{dil. H}_2\text{SO}_4 \longrightarrow$ no reaction {Cu is below H in electrochemical series}

Properties : (i) It is crystallised as $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

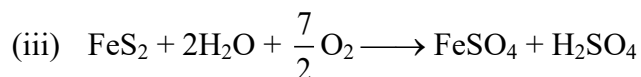
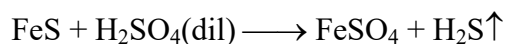


(iii) Revision with all others reagent

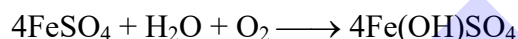
IRON COMPOUNDS**FeSO₄·7H₂O**

Preparation:→ (i) Scrap Fe + H₂SO₄ → FeSO₄ + H₂↑
(dil.)

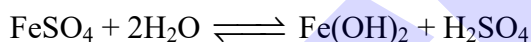
(ii) From Kipp's waste



Properties:→ (i) It undergoes aerial oxidation forming basic ferric sulphate

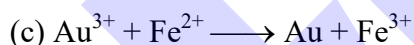
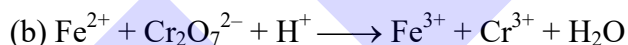


(iii) Aq. solution is acidic due to hydrolysis



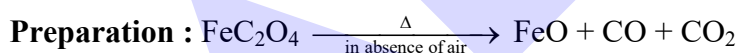
weak base

(iv) It is a reducing agent

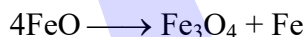
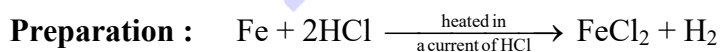


white ppt.

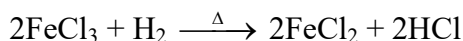
(v) It forms double salt. Example (NH₄)₂SO₄·FeSO₄·6H₂O

FeO(Black)

Properties : It is stable at high temperature and on cooling slowly disproportionates into Fe₃O₄ and iron

**FeCl₂**

OR

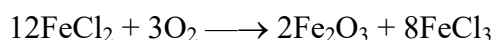


Properties :→ (i) It is deliquescent in air like FeCl₃

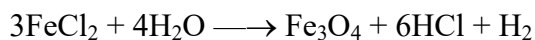
(ii) It is soluble in water, alcohol and ether also because it is sufficiently covalent in nature

(iii) It volatilises at about 1000°C and vapour density indicates the presence of Fe₂Cl₄. Above 1300°C density becomes normal

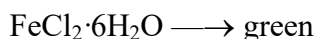
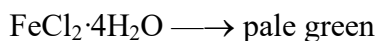
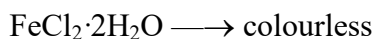
- (iv) It oxidises on heating in air



- (v)
- H_2
- evolves on heating in steam



- (vi) It can exist as different hydrated form

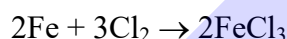


Ferric Chloride (FeCl_3)

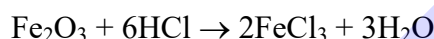
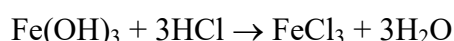
This is the most important ferric salt. It is known in anhydrous and hydrated forms. The hydrated form consists of six water molecules, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$.

Preparation :

- (i) Anhydrous ferric chloride is obtained by passing dry chlorine gas over heated iron fillings. The vapours are condensed in a bottle attached to the outlet of the tube.



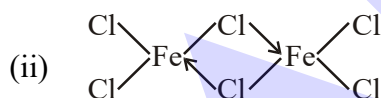
- (ii) $\text{Fe}_2(\text{CO}_3)_3 + 6\text{HCl} \rightarrow 2\text{FeCl}_3 + 3\text{H}_2\text{O} + 3\text{CO}_2$



The solution on evaporation and cooling deposits yellow crystals of hydrated ferric chloride. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$

Properties

- (i) $\text{Fe}_2\text{Cl}_6 \xrightleftharpoons{750^\circ\text{C}} 2\text{FeCl}_3 \xrightleftharpoons{\text{Above } 750^\circ\text{C}} 2\text{FeCl}_2 + \text{Cl}_2$



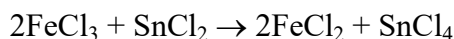
- (iii) It dissolves in water. The solution is acidic in nature due to its hydrolysis as shown below :



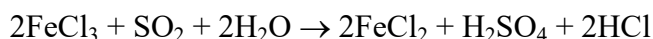
- (iv) $\text{FeCl}_3 + 6\text{NH}_3 \rightarrow \text{FeCl}_3 \cdot 6\text{NH}_3$

- (v) Ferric chloride acts as an oxidising agent.

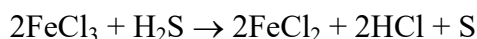
- (a) It oxidises stannous chloride to stannic chloride.



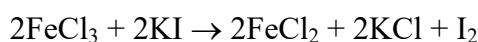
- (b) It oxidises SO_2 to H_2SO_4



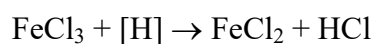
- (c) It oxidises H_2S to S



- (d) It liberates iodine from KI.



(e) Nascent hydrogen reduces FeCl_3 into FeCl_2



(vi) $\text{FeCl}_3 + 3\text{NH}_4\text{OH} \rightarrow \text{Fe}(\text{OH})_3 + 3\text{NH}_4\text{Cl}$

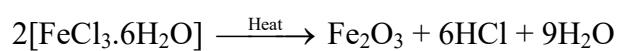
(vii) $\text{FeCl}_3 + \text{NH}_4\text{SCN} \rightarrow \text{Fe}(\text{SCN})\text{Cl}_2 + \text{NH}_4\text{Cl}$

or $\text{FeCl}_3 + 3\text{NH}_4\text{SCN} \rightarrow \text{Fe}(\text{SCN})_3 + 3\text{NH}_4\text{Cl}$

(viii) $4\text{FeCl}_3 + 3\text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Fe}_4[\text{Fe}(\text{CN})_6]_3 + 12\text{KCl}$

Prussian blue
(Ferri ferrocyanide)

(ix) On heating hydrated ferric chloride $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, anhydrous ferric chloride is not obtained. It is changed to Fe_2O_3 with evolution of H_2O and HCl .



Hydrated ferric chloride may be dehydrated by heating with thionyl chloride.

